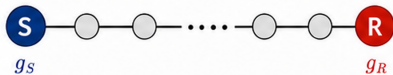


(a) Microscopic bus

Open XX chain with local endpoint couplings



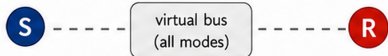
Microscopic Hamiltonian

$$\begin{aligned}\hat{H} &= \hat{H}_{SR} + \hat{H}_B + \hat{V}, \\ \hat{H}_B &= \frac{\omega_B}{2} \sum_{\ell=1}^L \hat{\sigma}_{\ell}^z + J_B \sum_{\ell=1}^{L-1} (\hat{\sigma}_{\ell}^+ \hat{\sigma}_{\ell+1}^- + \text{H.c.}), \\ \hat{V} &= g_S (\hat{\sigma}_S^+ \hat{\sigma}_1^- + \text{H.c.}) + g_R (\hat{\sigma}_R^+ \hat{\sigma}_L^- + \text{H.c.}).\end{aligned}$$

● S: sender qubit ● R: receiver qubit ○ chain spin

(b) All bus modes dispersive

Eliminate all bus modes



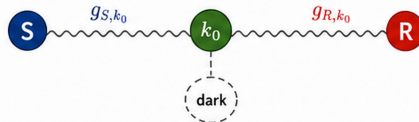
Endpoint-only effective Hamiltonian

$$\begin{aligned}\hat{H}_{SR}^{\text{disp}} &= \sum_{\nu=S,R} \frac{\tilde{\omega}_{\nu}}{2} \hat{\sigma}_{\nu}^z + J_{SR} (\hat{\sigma}_S^+ \hat{\sigma}_R^- + \text{H.c.}), \\ J_{SR} &= \frac{1}{2} \sum_k g_{S,k} g_{R,k}^* \left[\frac{1}{\Delta_{S,k}} + \frac{1}{\Delta_{R,k}} \right].\end{aligned}$$

- All appreciably coupled modes are off resonant: $|g_{\nu,k}/\Delta_{\nu,k}| \ll 1$.
- Bus occupation is perturbatively small, $P_B = \mathcal{O}(|g/\Delta|^2)$.
- Transfer is slow and mediated by a coherent signed sum over virtual modes.

(c) One isolated bus mode retained

Bright endpoint state couples to k_0 ; the orthogonal state is dark with respect to k_0 .



Selected-mode retained Hamiltonian

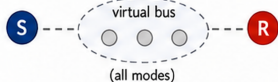
$$\begin{aligned}\hat{H}_{S k_0 R}^{\text{res}} &= \tilde{\omega}_S \hat{n}_S + \tilde{\epsilon}_{k_0} \hat{n}_{k_0} + \tilde{\omega}_R \hat{n}_R \\ &+ (g_{S,k_0} \hat{\sigma}_S^+ \hat{b}_{k_0} + g_{R,k_0} \hat{\sigma}_R^+ \hat{b}_{k_0} + \text{H.c.}),\end{aligned}$$

- k_0 is near resonant and spectrally isolated.
- The retained mode becomes physically populated and carries loss and memory.
- The dark endpoint combination decouples from k_0 , but is not generically immune to all noise.

(d) Choose the retained structure

All coupled modes dispersive

Eliminate all modes

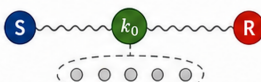


Endpoint-only Hamiltonian

- slow exchange;
- small virtual bus occupation;
- best when all relevant $|\Delta_{\nu,k}|$ are large.

One isolated pole

Retain k_0



Three-component model: S, k_0, R

- faster selected-mode transfer;
- real occupation of k_0 ;
- loss and memory remain explicit.

Several nearby poles

Retain a mode cluster $\mathcal{C}$



Multimode retained model

- captures spectral crowding;
- captures dispersion and delays;
- endpoint-only elimination fails.

Unknown, excited, or thermal bus

Change the encoding or use a finite-time channel map



- The polarized one-excitation reduction no longer applies.
- Initialization-free transfer may require a different encoding or endpoint protocol.
- A thermal finite bus is not automatically a Markovian reservoir.

← All relevant $|\Delta_{\nu,k}|$ large

→ One isolated denominator small

→ Several denominators small

Increasing departure from the fully dispersive spectral regime

Preparation / encoding branch — not part of the spectral axis